A Scalable Heisenberg Spin–Chain Model for a Phenalenyl–Triangulene Nanographene Polymer

1 From the nanographene geometry to a spin lattice

The polymer is a linear sequence of N identical monomers with the repeating motif $P-T-P$, connected in series under open boundary conditions (OBC),

$$P-T-P-\left[P-T-P\right]_{N-1}.$$

Each aromatic building block carries a well-defined low-energy spin degree of freedom of topological (π–electron) origin:

- a *phenalenyl* (P) is an alternant radical with sublattice imbalance 1, i.e. a single $S=\frac{1}{2}$ moment;
- a *[3]triangulene* (T) has sublattice imbalance 2 and hosts an $S=1$ moment. We represent it by two $S=\frac{1}{2}$ sites, T_a and T_b, locked into a triplet by a strong *ferromagnetic* intra-unit exchange $J_T<0$.

Consequently every monomer maps onto four $S=\frac{1}{2}$ nodes ordered as $[\,P,\,T_a,\,T_b,\,P\,]$, and the whole polymer maps onto a chain of

$$n=4N \tag{1}$$

spin-$\frac{1}{2}$ sites, indexed $i=0,\ldots,4N-1$. The magnetic couplings are inherited from the underlying mean-field Hubbard / second-order exchange description of the π system and take three values: the antiferromagnetic phenalenyl–phenalenyl coupling J_{PP}, the antiferromagnetic triangulene–phenalenyl coupling J_{TP}, and the ferromagnetic intra-triangulene coupling J_T.

2 Heisenberg Hamiltonian

With the convention $H=\sum_{\langle ij\rangle} J_{ij}\,\mathbf{S}_i\!\cdot\!\mathbf{S}_j$ and $\mathbf{S}_i=\frac{1}{2}\boldsymbol{\sigma}_i$, the parametric Hamiltonian of the $4N$-site chain reads

$$\boxed{H \;=\; \sum_{m=0}^{N-1}\Big[\,J_{TP}\big(\mathbf{S}_{4m}\!\cdot\!\mathbf{S}_{4m+1}+\mathbf{S}_{4m+2}\!\cdot\!\mathbf{S}_{4m+3}\big)+J_T\,\mathbf{S}_{4m+1}\!\cdot\!\mathbf{S}_{4m+2}\Big] \;+\; J_{PP}\sum_{m=0}^{N-2}\mathbf{S}_{4m+3}\!\cdot\!\mathbf{S}_{4m+4}} \tag{2}$$

The first sum runs over the three intra-monomer bonds ($J_{TP}\to J_T\to J_{TP}$) of each unit, while the second sum provides the $N-1$ inter-monomer J_{PP} links; in total there are $3N+(N-1)=4N-1$ bonds, as required for a linear chain with OBC. In the parametrization used here,

$$J_{PP}=38\text{ meV},\qquad J_{TP}=40\text{ meV},\qquad J_T=-500\text{ meV}. \tag{3}$$

Since $|J_T|\gg J_{TP},J_{PP}$, each (T_a,T_b) pair is effectively frozen into an $S=1$ object, and the model reduces to a mixed $\frac{1}{2}$–1–$\frac{1}{2}$ antiferromagnetic spin chain.

3 Exact diagonalization and Hilbert-space scaling

The model in Eq. (??) is $SU(2)$ symmetric and one dimensional, so exact diagonalization (ED) is the method of choice: it yields the numerically exact many-body ground state and low-lying spectrum, resolves total spin S and S_z, and gives the singlet–triplet spin gap directly, without any mean-field or variational bias. We implement it with `QuTiP` by building each spin operator as a tensor product of Pauli matrices and assembling H bond by bond.

The price is the exponential growth of the Hilbert space,

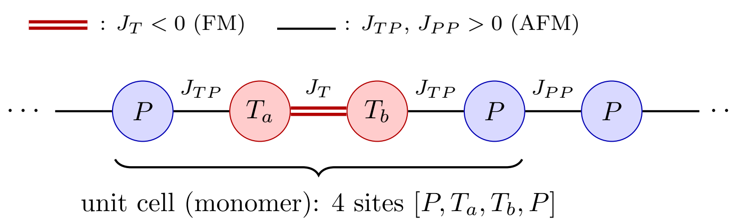
$$\dim \mathcal{H} \;=\; 2^{n} \;=\; 2^{4N}, \tag{4}$$

e.g. $2^{12} = 4096$ for $N = 3$ and $2^{16} = 65536$ for $N = 4$. Full dense diagonalization is therefore feasible only up to $n \sim 14$–16 sites; beyond that one exploits the conservation of total S_z (which block-diagonalizes H) together with sparse Lanczos/Arnoldi solvers to extract only the lowest eigenpairs, extending ED to roughly $n \sim 18$–20 sites.

By the Lieb–Mattis theorem, the bipartite chain with balanced sublattice spins ($S_A = S_B$) has a total-singlet ($S = 0$) ground state for every N; its local magnetization $\langle S_z^i \rangle$ vanishes identically by symmetry, and the intrinsic magnetic structure is instead revealed by the $S_z = +1$ member of the lowest $S = 1$ multiplet.

4 Conceptual 1D scheme

Figure below shows the repeating unit cell of spin nodes and the three exchange constants.



The two red nodes T_a, T_b tied by the double (ferromagnetic) J_T bond form the effective $S = 1$ triangulene; the blue P nodes are the $S = \frac{1}{2}$ phenalenyls. Translating the boxed unit cell N times and truncating at both ends produces the finite $4N$-site chain solved by ED.